ESP32-S3-BASED SMART ULTRASONIC RADAR SYSTEM FOR REAL-TIME OBJECT DETECTION AND DISTANCE MONITORING

INTERNET OF THINGS PROJECT REPORT SUBMITTED IN
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PROJECT WORK

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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QUALITY POLICY

To establish a system of Quality Enhancement, which would on a continuous basis evaluate and enhance the quality of teaching – learning, research and extension activities of the institution, leading to improvements in all processes, enabling the institution to attain excellence.

INSTITUTE VISION

To be recognized as a premier institution, grooming students into globally acknowledged engineering professionals.

INSTITUTE MISSION

- Providing outcome and value based engineering education.
- Nurturing research and entrepreneurial culture.
- Enabling students to be industry ready and fulfil their career aspirations.
- Grooming students through behavioral and leadership training programs.

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DEPARTMENT VISION

To groom students into futuristic and globally competent Electronics and Communication engineering professionals.

DEPARTMENT MISSION

- To impart quality education with moral and ethical values to develop competent engineers, leaders and successful entrepreneurs
- To establish state-of-art infrastructure and provide opportunities to update on emerging tools and technologies
- To empower the faculty towards excellence in teaching – learning, consultancy, research and development activities
- To foster socially relevant and industry-oriented innovation among students.

PROGRAM EDUCATIONAL OBJECTIVES

PEO1: Pursue career in multinational organizations, research organizations and core industries, higher studies at premier institutions and establish start-ups.

PEO2: Acquire core competencies in Electronics and Communication Engineering and exposure to latest Electronic Design Automation (EDA) tools.

PEO3: Exhibit professional skills and collaborative work experience.

PROGRAM OUTCOMES

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES

PSO1: Interpret and Design Electronic systems using Internet of Things, VLSI Technology and Efficient signal processing algorithms

PSO2: Apply knowledge to solve challenges in Communication Systems and Networks

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ABSTRACT

The detection of objects and the measurement of their distance and direction are important in applications such as obstacle detection, surveillance, robotics, and automated monitoring systems. Conventional distance-measuring devices generally provide information about an object in a single direction, which limits their ability to monitor a wider area. To overcome this limitation, this project presents a **Smart Ultrasonic Radar System Using ESP32-S3**, designed to detect nearby objects and display their approximate distance and angular position in real time.

The proposed system consists of an ESP32-S3 microcontroller, an HC-SR04 ultrasonic sensor, a servo motor, and an ST7735 TFT display. The ultrasonic sensor is mounted on a servo motor using a connecting rod and a 3D-printed holder, enabling the sensor to rotate across a defined angular range. The HC-SR04 sensor measures the distance to an object by transmitting ultrasonic waves and calculating the time taken for the reflected waves to return. The ESP32-S3 processes the measured distance and the corresponding servo angle to determine the approximate position of a detected object.

The measured information is displayed on the TFT screen through a radar-style graphical interface. The interface includes a semicircular radar grid, range indicators, a rotating sweep line, and numerical displays for the scanning angle and measured distance. Detected objects are represented using red markers, while the radar sweep is displayed in green to distinguish the scanning direction from detected targets. The system is developed using the PlatformIO development environment with the Arduino framework, and a 3D-printed mechanical structure is designed to support the display, servo motor, and ultrasonic sensor assembly.

The developed prototype demonstrates the integration of embedded programming, ultrasonic distance measurement, servo-based mechanical scanning, graphical display interfacing, and 3D-printed mechanical components. The system provides a compact and low-cost approach to visualizing object positions within a scanning region and can serve as a foundation for further development in robotics, obstacle detection, and automated monitoring applications.

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CHAPTER 1
INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Object detection and distance measurement are important in applications such as robotics, obstacle detection, and automated monitoring. Conventional distance-measuring systems often monitor objects in a single direction, limiting their ability to scan a wider area. The integration of ultrasonic sensors, servo motors, and microcontrollers provides an effective approach to detecting objects at different angles. This project develops a Smart Ultrasonic Radar System using ESP32-S3 to scan the surrounding area and display the approximate distance and angular position of detected objects in real time.

1.2 PROBLEM STATEMENT

Conventional distance-measuring systems provide limited directional coverage and may require manual movement to detect objects at different positions. Therefore, there is a need for a low-cost object detection system capable of scanning the surrounding area at different angles, measuring the distance to detected objects, displaying the scanning angle and distance in real time, visualizing detected objects using a radar-style interface, and providing a compact hardware implementation.

1.3 CORE OBJECTIVE OF THE PROJECT

The main objective of this project is to develop a Smart Ultrasonic Radar System using ESP32-S3 for scanning the surrounding area, detecting objects, measuring their distance, and displaying their approximate angular position on a TFT display.

Specific objectives

- To measure object distance using the HC-SR04 ultrasonic sensor.
- To rotate the sensor using a servo motor.
- To process sensor readings using the ESP32-S3 microcontroller.
- To display radar graphics using an ST7735 TFT display.
- To indicate detected objects using red markers.
- To display the scanning angle and measured distance.
- To develop a compact mechanical assembly using 3D-printed components.

1.4 SCOPE AND SIGNIFICANCE

The system is designed as a prototype for ultrasonic object detection and directional scanning. It can be extended for applications such as robotics and obstacle detection, automated monitoring systems, educational embedded systems projects, basic object-position visualization, and prototype navigation systems. The project demonstrates the integration of ultrasonic sensing, servo-based scanning, embedded programming, graphical visualization, and 3D-printed mechanical components in a low-cost radar system.

CHAPTER 2

LITERATURE SURVEY

2.1 REVIEW OF EXISTING SYSTEMS AND SOLUTIONS

Object detection and distance measurement systems are widely used in robotics, automation, and monitoring applications. Conventional ultrasonic distance-measuring systems use sound-wave reflections to estimate the distance between a sensor and an object. Servo-based scanning systems extend this approach by rotating the sensor to measure objects at different angular positions. Microcontrollers process the sensor readings and can display the measured information through graphical interfaces. Radar-style displays provide a visual representation of the scanning direction and detected objects, making the system easier to interpret.

2.2 LIMITATIONS OF EXISTING SOLUTIONS

Some conventional object detection and distance measurement systems have the following limitations:

- Limited detection coverage when the sensor faces only one direction.
- Manual movement may be required to inspect different positions.
- Basic distance sensors may not provide a visual representation of object direction.
- Commercial radar and scanning systems can be costly for simple applications.
- Mechanical alignment and sensor readings can affect detection reliability.

The proposed project addresses these limitations through a low-cost prototype combining an ESP32-S3, an HC-SR04 ultrasonic sensor, a servo motor, and a TFT display.

2.3 COMPARATIVE STUDY ON ALTERNATIVE SOLUTIONS

Parameter	Conventional Distance Sensor	Proposed Ultrasonic Radar
Detection direction	Usually one direction	Servo-based angular scanning
Distance measurement	Supported	Supported using HC-SR04
Angular position display	Usually not available	Displayed on TFT
Visual interface	Basic or unavailable	Radar-style graphics
Controller	Varies by system	ESP32-S3
Implementation cost	Varies	Low-cost prototype

Table 2.1: Comparative Study of Conventional and Proposed Systems

CHAPTER 3

PROPOSED SOLUTION

3.1 Proposed System Overview

The proposed Smart Ultrasonic Radar System uses an ESP32-S3 microcontroller to control an HC-SR04 ultrasonic sensor mounted on a servo motor. The servo rotates the sensor across a defined angular range to detect objects at different positions. The ultrasonic sensor measures the distance by transmitting ultrasonic waves and calculating the time taken for the reflected waves to return.

The ESP32-S3 processes the sensor readings and displays the scanning angle and measured distance on an ST7735 TFT display. The radar interface consists of a semicircular grid, range indicators, a moving sweep line, and red markers representing detected objects. A 3D-printed structure supports the display, servo motor, and ultrasonic sensor, providing a compact and organized hardware assembly.

3.2 SYSTEM ARCHITECTURE / BLOCK DIAGRAM

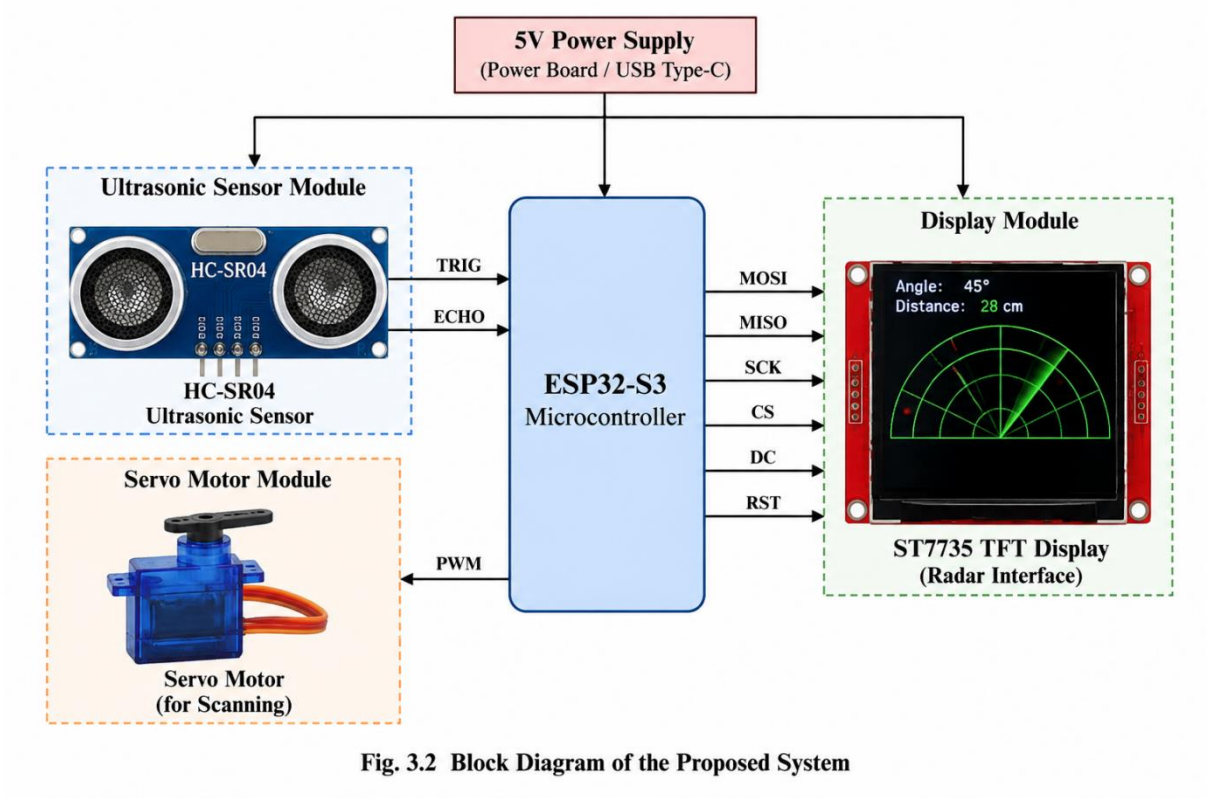


Fig. 3.2 Block Diagram of the Proposed System

Figure 3.1: System Architecture / Block Diagram

3.3 WORKFLOW / PROCESS FLOW

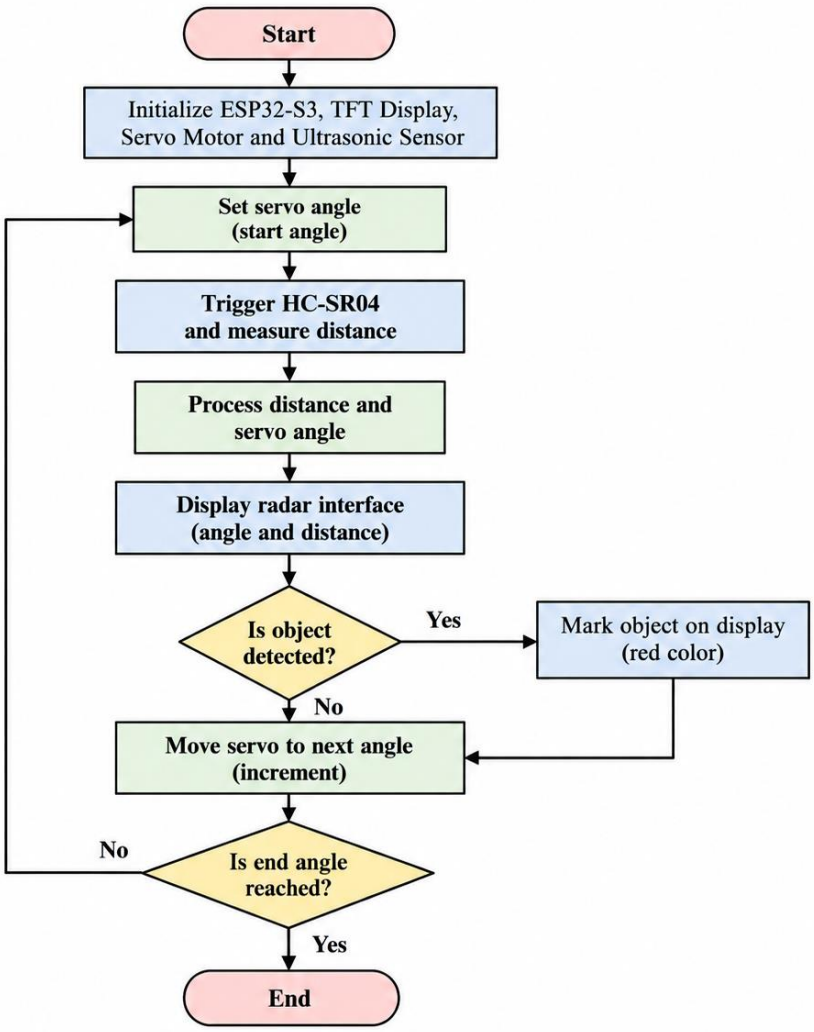


Fig. 3.3 Flowchart of the Proposed System

3.4 FUNCTIONAL REQUIREMENTS

The system shall:

- Rotate the ultrasonic sensor using a servo motor.
- Measure object distance using the HC-SR04 ultrasonic sensor.
- Process sensor readings using the ESP32-S3 microcontroller.
- Display radar graphics on the ST7735 TFT display.
- Display the scanning angle and measured distance.
- Indicate detected objects using red markers.

3.5 NON-FUNCTIONAL REQUIREMENTS

· Low-cost hardware implementation.

· Real-time operation and display updates.

· Compact and portable design.

· Simple hardware interfacing and maintenance.

· Stable servo operation and reliable distance measurement.

· Expandability for future improvements.

CHAPTER 4

HARDWARE DESCRIPTION

4.1 HARDWARE COMPONENTS USED

S. No.	Component	Quantity	Purpose
1	ESP32-S3 Development Board	1	Main controller for processing sensor data and controlling the system
2	HC-SR04 Ultrasonic Sensor	1	Measures the distance to detected objects
3	SG90 Servo Motor	1	Rotates the ultrasonic sensor for angular scanning
4	ST7735 TFT Display (1.8-inch)	1	Displays radar graphics, scanning angle, and distance
5	5V Power Supply / Power Board	1	Supplies power to the system and servo motor
6	Resistors (Voltage Divider)	2	Reduces the HC-SR04 ECHO voltage to a safe level for the ESP32-S3
7	Jumper Wires	As required	Provides electrical connections between components
8	Breadboard	1	Supports temporary circuit assembly and testing
9	3D-Printed Mounting Components	As required	Holds the display, servo motor, and ultrasonic sensor assembly

Table 4.1: Hardware Components Used

4.1.1 ESP32-S3

The ESP32-S3 acts as the main controller of the radar system. It processes the ultrasonic sensor readings, controls the servo motor, calculates the distance to detected objects, and updates the radar graphics on the TFT display.

4.1.2 HC-SR04 Ultrasonic Sensor

The HC-SR04 ultrasonic sensor is used to measure the distance to nearby objects by transmitting ultrasonic waves and detecting their reflections. The time taken for the reflected waves to return is used to calculate the distance.

4.1.3 Servo Motor

The servo motor rotates the ultrasonic sensor across a defined angular range. It receives control signals from the ESP32-S3, allowing the sensor to scan different directions and detect objects at different angular positions.

4.1.4 ST7735 TFT Display

The ST7735 TFT display provides a graphical interface for the radar system. It displays the radar grid, scanning line, detected objects, scanning angle, and measured distance.

4.1.5 Power Supply

The power supply provides the required voltage to the ESP32-S3 and peripheral components. An external 5V supply is used for the servo motor to support its power requirements.

4.1.6 Voltage Divider

The voltage divider reduces the HC-SR04 ECHO signal voltage to a level suitable for the ESP32-S3 GPIO input, helping protect the microcontroller from excessive voltage.

4.1.7 3D-Printed Components

The 3D-printed components include the sensor holder, connecting rod, and display holder. These components provide mechanical support and enable the ultrasonic sensor to rotate with the servo motor.

4.2 HARDWARE INTERFACING AND PIN CONFIGURATION

ESP32 Transmitter

S. No.	Component	Pin / Signal	ESP32-S3 GPIO Pin
1	HC-SR04 Ultrasonic Sensor	TRIG	GPIO 7
2	HC-SR04 Ultrasonic Sensor	ECHO	GPIO 16
3	Servo Motor	Signal	GPIO 17
4	ST7735 TFT Display	CS	GPIO 4
5	ST7735 TFT Display	DC (A0)	GPIO 15
6	ST7735 TFT Display	RESET	GPIO 9
7	ST7735 TFT Display	MOSI (SDA)	GPIO 11
8	ST7735 TFT Display	SCK	GPIO 12

Note: The HC-SR04 ECHO pin is connected through a voltage divider to protect the ESP32-S3's 3.3 V GPIO input. The servo motor is powered using an external 5 V supply, with the power supply ground connected to the ESP32-S3 ground.

4.3 CIRCUIT DIAGRAM

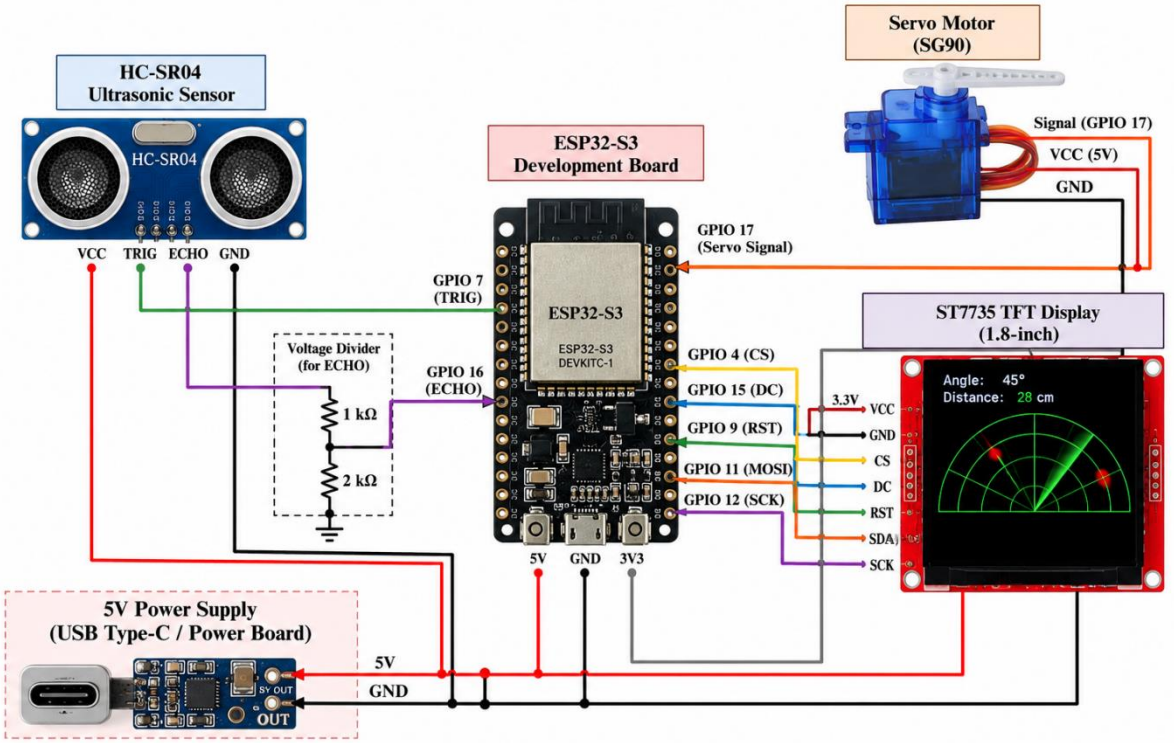


Fig. 4.2 Circuit Diagram of the Proposed System

4.4 HARDWARE SETUP AND IMPLEMENTATION

The Smart Ultrasonic Radar System operates by rotating the HC-SR04 ultrasonic sensor using a servo motor to scan objects at different angular positions. The ESP32-S3 controls the servo motor and triggers the ultrasonic sensor to transmit ultrasonic waves. When the waves reflect from an object, the sensor receives the echo signal, and the ESP32-S3 calculates the distance based on the echo time.

The measured distance and scanning angle are processed by the ESP32-S3 and displayed on the ST7735 TFT display. The display presents a radar grid, a scanning line, and red markers to indicate detected objects. This process repeats continuously to scan the surrounding area and provide visual information about the position and distance of nearby objects.





Figure 4.3: WORKING SIMULATION

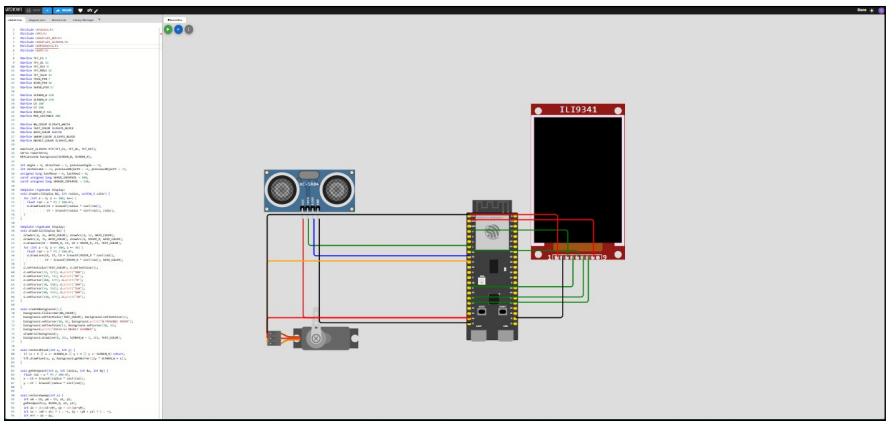


Figure 4.4: WOKWI SIMULATION RESULT

CHAPTER 5

SOFTWARE DESIGN

5.1 PROGRAMMING ENVIRONMENT / IDE

· PlatformIO IDE

· Visual Studio Code

· Arduino Framework

· ESP32-S3 Development Environment

· TFT_eSPI Library

5.2 COMMUNICATION PROTOCOLS

SPI (Serial Peripheral Interface)

Used between the ESP32-S3 and the ST7735 TFT display for transferring display data.

GPIO (General-Purpose Input/Output)

Used for interfacing the ESP32-S3 with the HC-SR04 ultrasonic sensor and servo motor.

PWM (Pulse Width Modulation)

Used to control the servo motor's angular position through signals generated by the ESP32-S3.

Ultrasonic Echo Communication

The HC-SR04 transmits ultrasonic waves and receives reflected waves. The ESP32-S3 measures the echo duration to calculate the distance to an object.

5.3 Radar User Interface Design

The radar interface is developed using the ST7735 TFT display and the TFT_eSPI library.

The interface provides:

- Radar grid and range indicators
- Rotating radar sweep line
- Red markers indicating detected objects
- Scanning angle display
- Measured distance display
- Continuous graphical updates

- Hazard explanation
- Firebase connection status

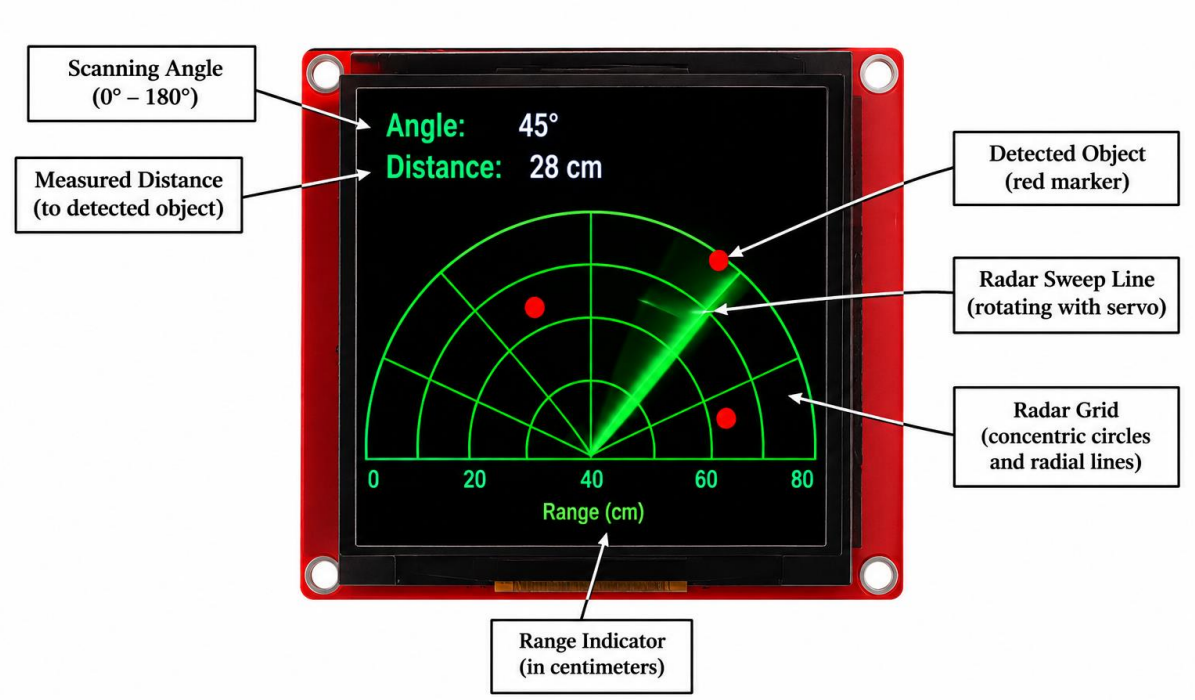


Figure 5.1: Radar Display Interface (ST7735 TFT Display)

5.4 Radar Display Implementation

The radar display is implemented using the ESP32-S3 and the ST7735 TFT display. The ESP32-S3 processes the ultrasonic sensor readings and updates the radar graphics according to the scanning angle and measured distance. The radar interface displays a semicircular grid with range indicators and a moving sweep line. When an object is detected, a red marker represents its position on the radar screen. The scanning angle and measured distance are also displayed to provide information about nearby objects. The servo motor rotates the ultrasonic sensor across the scanning range, allowing the system to examine different directions. The display is continuously refreshed to represent the scanning process visually

CHAPTER 6

IMPLEMENTATION AND INTEGRATION

6.1 Step-by-Step Implementation

The implementation of the proposed Smart Ultrasonic Radar System was carried out systematically by integrating the ultrasonic sensor, servo motor, ESP32-S3 microcontroller, TFT display, and power supply.

1. ESP32-S3 Hardware Configuration: The ESP32-S3 was configured as the main controller and programmed using the PlatformIO development environment with the Arduino framework.

2. Ultrasonic Sensor Integration: The HC-SR04 ultrasonic sensor was connected to the ESP32-S3 to measure the distance to nearby objects. The TRIG and ECHO pins were configured for ultrasonic signal transmission and reception.

3. Servo Motor Integration: The servo motor was connected to GPIO 17 of the ESP32-S3. It was programmed to rotate the ultrasonic sensor across a defined angular range for scanning different directions.

4. TFT Display Integration: The ST7735 TFT display was interfaced with the ESP32-S3 using SPI communication. The display was configured to render radar graphics and sensor information.

5. Power Supply Configuration: An external 5V power supply was used to power the servo motor. A common ground was established between the power supply and ESP32-S3 to support proper signal communication.

6. Voltage Divider Implementation: A voltage divider was connected between the HC-SR04 ECHO output and GPIO 16 to reduce the signal voltage to a level suitable for the ESP32-S3.

7. Servo Scanning Implementation: The ESP32-S3 generated control signals to rotate the servo motor through the programmed scanning angles. The scanning angle was updated during the operation.

8. Distance Measurement: The ESP32-S3 triggered the HC-SR04 sensor and measured the duration of the returned echo signal to calculate the distance to an object.

9. Radar Graphics Development: The TFT display was programmed to render a semicircular radar grid, range indicators, and a scanning line representing the current scanning direction.

10. Object Detection Visualization: The measured distance and scanning angle were processed to represent detected objects using red markers on the radar display.

11. System Integration: The ESP32-S3, ultrasonic sensor, servo motor, TFT display, and power supply were integrated into a single prototype. The hardware connections and software operation were checked.

12. 3D-Printed Assembly: A 3D-printed structure consisting of a sensor holder, connecting rod, and display holder was used to support and organize the system components.

13. Display and Scanning Testing: The radar graphics and servo scanning operation were tested to verify the display updates and movement of the scanning mechanism.

14. Distance Measurement Testing: The ultrasonic sensor was tested with objects placed at different distances. The readings were examined to evaluate the sensor's response during scanning.

15. Complete System Testing: The integrated prototype was operated to observe the scanning angle, distance display, radar sweep, and object markers on the TFT screen

6.2 Hardware and Software Integration

The complete system integrates **ultrasonic distance measurement, servo-controlled scanning, microcontroller processing, and graphical visualization** into a single radar prototype. The ESP32-S3 acts as the central controller, coordinating the HC-SR04 ultrasonic sensor, servo motor, and ST7735 TFT display.

The HC-SR04 sensor measures the distance to nearby objects by transmitting ultrasonic waves and receiving their reflections. The servo motor rotates the sensor to scan different angular positions. The ESP32-S3 processes the echo duration and scanning angle to obtain distance information and determine where detected objects should appear on the radar interface.

The ST7735 TFT display presents the radar grid, scanning line, range indicators, and red markers representing detected objects. The scanning angle and measured distance are also displayed to provide visual information about the surrounding area.

The hardware components are supported by an external 5V power supply for the servo motor, with a common ground shared by the system components. The 3D-printed sensor holder, connecting rod, and display holder provide mechanical support for the assembled prototype.

The integration of these hardware and software components demonstrates the implementation of an embedded radar system using the ESP32-S3. The prototype combines servo movement, ultrasonic sensing, distance processing, and graphical display to visualize objects at different scanning angles.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 TESTING AND VALIDATION

The Smart Ultrasonic Radar System was tested to evaluate servo motor movement, ultrasonic distance measurement, and radar graphics displayed on the ST7735 TFT display.

Servo Motor Testing

The servo motor was tested to verify its movement across the programmed scanning angles. The ESP32-S3 generated control signals to rotate the sensor and change its scanning direction.

Ultrasonic Sensor Testing

The HC-SR04 ultrasonic sensor was tested to measure the distance to nearby objects. The sensor's response was examined by observing the echo signal and the calculated distance values.

TFT Display Testing

The ST7735 TFT display was tested to verify the rendering of the radar grid, scanning line, range indicators, and numerical angle and distance information.

Radar Visualization Testing

The radar interface was checked to verify the display of detected-object markers and the association between scanning angles and measured distances.

7.2 HARDWARE SETUP AND RESULTS

The final prototype consists of an ESP32-S3 development board, HC-SR04 ultrasonic sensor, servo motor, ST7735 TFT display, external 5V power supply, and 3D-printed mounting components.

The ESP32-S3 acts as the main controller, while the servo motor rotates the ultrasonic sensor to scan different directions. The sensor measures the distance to objects, and the controller processes the readings to update the radar display.

The TFT display presents the radar graphics, including the semicircular grid, scanning line, detected-object markers, scanning angle, and distance information. The 3D-printed sensor holder, connecting rod, and display holder provide mechanical support for the prototype.

Figure 7.1: Final Hardware Prototype



Figure 7.2: Radar Display Output

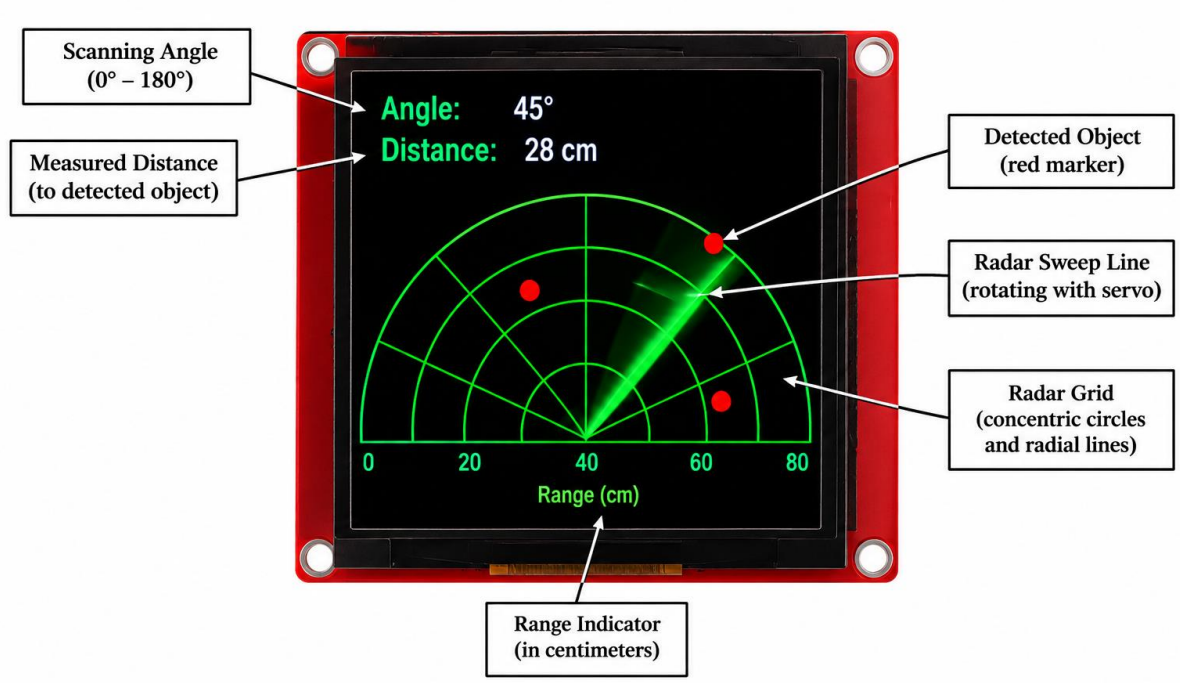


Figure 5.1: Radar Display Interface (ST7735 TFT Display)

7.3 SOFTWARE RESULTS

The software was developed using the Arduino framework in PlatformIO within Visual Studio Code. The program controls the servo motor, triggers the HC-SR04 ultrasonic sensor, processes echo signals, and updates the ST7735 TFT display.

The radar interface displays the scanning angle and measured distance along with the radar grid and sweep line. Red markers are used to represent detected objects on the display.

During testing, the TFT display successfully rendered the radar graphics, and the servo motor operated. However, the ultrasonic sensor produced repeated *NO ECHO* readings during debugging, indicating that reliable distance measurement had not yet been established. Therefore, further testing of the sensor connections, voltage divider, power supply, and echo signal is required to validate the complete radar detection function.

Example test values:

Table 7.1: Example Test Values

S. No.	Component / Function	Observation
1	ESP32-S3	Controller programmed using PlatformIO
2	TFT Display	Radar graphics displayed
3	Servo Motor	Servo movement observed
4	HC-SR04 Sensor	Repeated NO ECHO readings during testing
5	Radar Visualization	Radar grid and sweep line displayed
6	Distance Measurement	Reliable readings not yet confirmed

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION

The Smart Ultrasonic Radar System using ESP32-S3 was developed to demonstrate object scanning and distance visualization using an ultrasonic sensor, servo motor, and TFT display. The ESP32-S3 acts as the main controller, coordinating the servo motor movement, ultrasonic sensor operation, and radar graphics displayed on the ST7735 TFT display.

The system incorporates a servo-mounted HC-SR04 ultrasonic sensor to scan different angular positions. The radar interface displays a semicircular grid, scanning line, range indicators, scanning angle, and red markers representing detected objects. The 3D-printed sensor holder, connecting rod, and display holder provide mechanical support for the prototype.

During testing, the TFT display successfully rendered the radar graphics, and the servo motor operated. However, reliable ultrasonic distance measurement could not be confirmed because the sensor repeatedly produced NO ECHO readings. Therefore, further debugging and validation are required to achieve reliable object detection and distance measurement.

The project demonstrates the integration of embedded programming, ultrasonic sensing, servo motor control, graphical display interfacing, and 3D-printed mechanical components. With improvements to sensor reliability and system calibration, the prototype can serve as a foundation for low-cost radar-based object detection and obstacle monitoring applications.

8.2 FUTURE SCOPE

Future improvements to the proposed system can include:

- Improving ultrasonic sensor reliability and echo detection.
- Enhancing distance measurement accuracy through calibration.
- Improving servo motor stability and scanning smoothness.
- Adding multiple distance sensors to increase detection coverage.
- Implementing audible alerts using a buzzer when objects are detected nearby.
- Extending the scanning range and improving radar visualization.
- Adding wireless connectivity for remote monitoring.
- Developing object tracking and obstacle avoidance capabilities.
- Optimizing power consumption for portable operation.
- Improving the 3D-printed enclosure for better component protection and portability.
- Integrating the system with robotic platforms for navigation and obstacle detection.

CHAPTER 9

REFERENCES

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Annexure - I

PROJECT TITLE	**SMART ULTRASONIC RADAR SYSTEM USING ESP32-S3**
PROGRAM	B.E. ECE
STUDENT NAME	RAKESH R
REGISTRATION NUMBER	722824106126
NAME OF THE SUPERVISOR AND DESIGNATION	Dr. S. Ramalingam, AP/ECE
NAME OF THE SDG GOALS MAPPED	SDG 9: Industry, Innovation, and Infrastructure SDG 11: Sustainable Cities and Communities
SDG GOAL NUMBER	SDG 9,SDG 11
NAME OF THE TRL LEVEL	Technology Validated in Laboratory
MENTION THE TRL LEVEL	TRL 4

POs & PSOs Mapping (Put a tick mark in the mapped PO's & PSO's):

Program Outcomes											**Program Specific Outcomes**	
PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**
✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓

Supervisor
Dr. S. Ramalingam, AP/ECE

Annexure - II

VENUE AND EXPENDITURE STATEMENT FOR THE PROJECT WORK

Laboratory details where the project is carried out	
Software Details	· PlatformIO IDE · Visual Studio Code · Arduino Framework · TFT_eSPI Library · ESP32-S3 Development Environment

Details of the Component and Expenditure

S. No.	Name of Component	Quantity	Price / Unit (₹)	Amount (₹)
1	ESP32-S3 Development Board (N16R8)	1	950	950
2	HC-SR04 Ultrasonic Sensor	1	70	70
3	SG90 Servo Motor	1	120	120
4	ST7735 TFT Display (1.8-inch)	1	300	300
5	5V Power Supply / Power Board	1	100	100
6	Resistors for Voltage Divider	2	5	10
7	Jumper Wires	As required	50	50
8	Breadboard	1	60	60
9	3D-Printed Mounting Components	As required	300	300
	TOTAL EXPENDITURE			**₹1,960**

Signature of the student
RAKESH R

Supervisor
Dr. S. Ramalingam, AP/ECE